

A2 then A1 for a Free 1d Fermion: The Local- X Ansatz is Not Refuted; the UV Coefficient is IR-Independent

Robert W. McGwier
CTO, Cohere Technology Group
GitHub: [n4hy](#)

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Abstract

Jacobson 2016’s nonconformal step is a conjecture: $\delta\langle K \rangle = c\delta\langle T_{00} \rangle + \delta X$ with X a spacetime scalar. On a lattice the scalar is diagonal. The local ansatz is therefore a *tridiagonal* modular kernel.

A2, pre-registered: on a 1d staggered-mass fermion, vacuum interval of $L = 32$, $R(m) = \|K - K_{\text{tri}}\|_F / \|K\|_F$ obeys $R(m)/R(0) < 2$ for all $0 < mL \leq 8$. Measured ratios: 1.07, 1.16, 1.32 (solver) and 1.26, 1.69 (auditor, different N, L). C2 PASS. The mutation (drop the diagonal) fires: a local mass term is visible. [P] as a necessary-condition test on this field, not as a 4d diamond theorem.

A1, pre-registered: the UV log coefficient $\alpha = (S(2L) - S(L))/\log 2$ at $mL \leq 0.5$ stays within 20% of $\alpha(0) = 0.323 \approx 1/3$. Measured relative drifts 0.021 and 0.072. At $mL = 6.4$, α collapses (mutation). [P] for this 1d free fermion. Not a 4d area law.

Road A may continue for this field class. It still does not give FGHMV in de Sitter or a value of Λ .

1 A2

$K = \log((1 - C)/C)$ from the restricted correlator of the ground state of $H_{i,i+1} = -1$, $H_{ii} = m(-1)^i$. Local ansatz = diagonal + nearest neighbour. Remainder of range ≥ 2 cannot be a scalar X .

mL	R	$R/R(0)$	in window
0	0.160	1	—
1.6	0.171	1.07	yes
3.2	0.186	1.16	yes
6.4	0.211	1.32	yes
12.8	0.239	1.49	no

C2 threshold was 2, locked before the run. The ratio grows. It does not cross 2 inside the window. The conjecture is *not refuted* here. It is not proved in 4d.

2 A1

$\alpha(0) = 0.323$. UV masses $mL \leq 0.32$ shift α by at most 7%. IR mass $mL = 6.4$ kills the log ($\alpha = 0.010$). Leading UV piece is IR-mass independent for this fermion. [P]

3 What this does not do

It does not prove A2 for a 4d diamond or for the Standard Model. It does not derive $\eta = 1/4G$. It does not select Λ . It does not produce FGHMV in de Sitter.

Admission. *A2 passed a necessary-condition test I could fail. A1 passed on the same field. I will not write that Jacobson's conjecture is a theorem in four dimensions.*